Generative AI Audit Log — MP 5

CIS 531/731 — Fall 2026 — Structured Streaming

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Assignment	MP 5: Data Analytics Pipelines (Structured Streaming)
Track	Track 2 — VDB & Trope Ontologies (no live ASR capture)
Tool audited	Claude (Anthropic) — conversational assistant
Period covered	19–20 October 2026
Deliverable	report.md, src/*.py, out/metrics.jsonl, notebooks/mp5_reconciliation.ipynb

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1. Scope and Policy Position

MP 5 Section 2 ("Initiating Your Draft — The AI Kit Procedure") directs students to use the Generative AI Kit for contract extraction, pipeline scaffolding, event-time calibration, fault injection, and run-report assembly. AI assistance on this milestone is therefore instructed rather than merely permitted, and this log records what it actually did. The same specification attaches a verification duty to each stage — "then verify every nullability choice yourself", "audit it for schema inference, missing watermarks, unbounded state, and non-idempotent sinks" — and that duty is where the substance of this entry lies.

I am disclosing the division of labour honestly, including the parts where the assistant did more than scaffold. Where a judgement in report.md is mine, I say so; where the assistant proposed something I adopted, I say that instead.

2. Assistance by Stage

Spec stage	What the assistant did	What I verified or decided
1. Contract extraction	Derived the StructType and an initial contract set from the MP 5 text (BoundedChunking, ts_start < ts_end, whitelist, provenance).	Reviewed each field's nullability against its downstream consumer; added IngestCausality after reasoning about what a corrupt producer clock does to the lateness distribution.
2. Pipeline scaffold	Wrote the readStream / withWatermark / groupBy(window(...)) skeleton with an explicit schema and per-query checkpoints.	Audited it as instructed. The audit found the three-valued-logic hole in the admission filter and the non-idempotency risk in update mode.
3. Event-time calibration	Implemented the percentile measurement and proposed ceil(p99 / 30s) as the rounding rule.	Chose p99 over max, and chose to measure on admitted records only. Both are argued in report.md.
4. Fault injection	Implemented os._exit(1) kill, restart, and the reconciliation harness.	Rejected the first reconciliation as invalid (see §3) and required it be rebuilt on order-independent invariants.
5. Run report	Drafted report.md from the measured JSON artifacts and assembled metrics.jsonl.	Every figure traced back to out/*.json. Re-ran run_all.sh from an empty directory to confirm reproducibility.

3. Defects Found by Running It, Not by Reading It

The spec's audit duty is the reason this submission is not the first thing that ran. Five defects were found and fixed, each by a measurement disagreeing with an expectation:

(a) array_remove returns null. violation_reasons() was first written as array_remove(array(...), None). Spark's array_remove returns null when the element to remove is null, so every violation reason came back null. The run reported 57 quarantined records and zero reasons; reconciling per-contract counts against the generator's manifest exposed it.

(b) Three-valued logic in the admission filter. A null in the conjunction yields null, and a null-gated row is admitted by neither where(gate) nor where(not gate) — it would vanish from every count. Fixed with coalesce(..., False) so the two branches partition the input exactly.

(c) The watermark was decorative. With files written in event-time order, the file source delivers records in roughly event-time order, so nothing ever arrives late and the job produces identical output with a 0 s watermark. The completeness-vs-latency trade-off could not be measured at all. Fixed by partitioning the fixture by ingest_ts, after which 372 of 1,000 records arrive out of order.

(d) FileStreamSource guarantees no arrival order. With every file present up front, unread files are taken in filesystem-listing order, so the drop set is not predictable from filenames. The first reconciliation could not account for 163 records and the engine's own drop counter disagreed with the per-window deficit by 129. I rejected that reconciliation rather than reporting it, and the replay was rebuilt as a staged feeder — a file that does not exist cannot be read — which made the accounting exact.

(e) StreamingQueryProgress has no batchDuration field. The spec names batchDuration; reading p["batchDuration"] returns None. The correct field is durationMs.triggerExecution. The first metrics log had an all-null batchDuration column.

4. What Was Not Resolved, and Is Not Hidden

One record of 943 (c000306) sits on the late-data filter's boundary: its window ends 14:16:00 against a reported watermark of 14:16:05, so the independent model predicts a drop and the engine admitted it. Four of five predicted drops are confirmed and the derived watermark matches the engine's reported value for all 40 non-empty batches, so the model is not broadly wrong. I considered adjusting the drop rule until the counts matched and did not, because that would be fitting a rule to a single observation. The check is asserted to within one record and the residual is stated in report.md and in out/reconciliation.json.

5. Uses Explicitly Not Made

AI was not used to fabricate, estimate, or interpolate any measured value: every number in report.md is emitted by code in src/ and stored in out/. It was not used to invent citations. The landing-zone fixture is synthetic and is disclosed as synthetic in report.md, in the generator's module docstring, and in landing_manifest.json ("synthetic": true) — it is a seeded stand-in for the MP 3 corpus snapshot, not AI-generated data presented as real. No reconciliation result was accepted without the harness exiting zero.

6. Attestation

I can explain and defend every design decision in report.md without assistance: why the watermark is p99 and not the maximum, why append mode is required rather than chosen, why the top-10 is computed at read time, why exactly-once needs both the checkpoint and the idempotent sink, and why the record accounting is a closed balance rather than a spot check. Where the assistant's contribution exceeded scaffolding, §2 and §3 say so.

Signed: _____ Date: _____

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